from PagerDuty and cached for only a few minutes. If an engineer goes off call, subsequent requests are immediately re-evaluated.

Over time, ACME learned that the toughest part of maintaining freshness isn't just setting cache durations—it's managing dependencies. A cached entity might combine attributes with very different lifetimes. Employment status may change once a year, but device posture can change hourly. ACME's enterprise-level signal processing system, introduced later in this chapter, handles these differences by tracking each signal's source, TTL, and validation method.

Freshness is never free. It involves balancing speed, reliability, and assurance. By considering it as a key design dimension rather than an afterthought, organizations like ACME can tune their authorization context to match the risk and performance needs of each system.

12.5 Transmission and integrity of signals

Trust signals are only as useful as they are trustworthy. Even after they've been gathered from their sources, they must still travel securely from the systems that generate them to the systems that use them. When a signal is corrupted or falsified in transit, the consequences range from bad authorization decisions and user frustration to a direct path for attackers to gain unauthorized access.

Signal transmission is the connective tissue of authorization context. Each hop—from the originating system to the Policy Information Point (PIP), then to the Policy Decision Point (PDP), and finally to the Policy Enforcement Point (PEP)—must maintain authenticity, integrity, and confidentiality.

12.5.1 How signals move

Signals reach the PDP in several ways:

- **Embedded in tokens.** Some signals, like device posture or authentication context, are packaged as signed tokens that travel with the access request. Their integrity is maintained through cryptographic